# NV-Center Optical Readout of Superconducting Bolometer Pixels: Architecture, Cryogenic Diamond Characterization, and Multi-Modal Sensing


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***Abstract*—We present a bolometric detector architecture that leverages nitrogen-vacancy (NV) center magnetometry to probe superconducting thin-film pixels, and report extensive cryogenic characterization of the NV-diamond sensing element from 11 K to 200 K. The detector concept exploits the thermal sensitivity of the Meissner effect: energy deposition from a particle heats a superconducting pixel past its critical temperature, collapsing Meissner screening and allowing an applied magnetic field to reach nearby NV centers. The resulting shift in optically detected magnetic resonance (ODMR) frequency produces a fluorescence plateau whose duration encodes the deposited energy. We describe the device architecture and present simulation results for a proof-of-principle NbN/Ge demonstrator targeting MeV-scale energy deposition. We report measurements of the zero-field splitting $D(T)$, transverse strain parameter $E(T)$, ODMR contrast, and spin-lattice relaxation time $T_1$ as functions of temperature, establishing the operating parameters required for detector design. Additionally, we demonstrate the platform's multi-modal sensing capability through real-time broadband Raman monitoring of structural phase transitions in amorphous solid water, observed serendipitously during cryostat operation. These results validate the cryogenic NV-diamond platform and outline the pathway to sub-eV threshold detection relevant for light dark matter searches.**



***Index Terms*—quantum sensing, NV centers, superconducting detectors, ODMR, bolometer, magnetometry, cryogenic diamond, amorphous ice**


## I. Introduction

Superconducting bolometers offer exceptional energy resolution for particle detection, but their readout—typically via transition-edge sensors (TES) or kinetic inductance detectors (KIDs)—requires complex cryogenic multiplexing using SQUID amplifiers and microwave feedlines [1]. This wiring overhead limits scalability and introduces thermal loads that constrain detector performance.

We propose a fundamentally different approach: using the optical magnetometry capabilities of nitrogen-vacancy (NV) centers in diamond to read out superconducting pixels with no electrical connection to the cold stage. NV centers provide shot-noise-limited magnetic field sensitivity across a wide dynamic range [2], [3], and their optical interface enables straightforward multiplexing via imaging. This architecture offers a pathway to large-area, densely pixelated detectors with sub-eV energy thresholds.

The operating principle exploits the temperature dependence of the London penetration depth $\lambda(T)$ in superconductors. Below $T_c$, Meissner screening excludes an applied magnetic field from the bulk. Energy deposition increases the local temperature, increasing $\lambda(T)$ and allowing field penetration. NV centers positioned near the superconducting surface detect this field change through shifts in their spin resonance frequencies.

Realizing this concept requires detailed understanding of NV-center behavior at cryogenic temperatures. In this work, we present comprehensive characterization of an ensemble NV-diamond sensor from 11 K to 200 K, including the temperature dependence of the zero-field splitting $D(T)$, transverse strain parameter $E(T)$, ODMR contrast, and spin-lattice relaxation time $T_1$. These measurements establish the operating envelope for detector design.

## II. Device Architecture

### A. Layer Stack and Geometry

The detector stack comprises (from top): a sapphire substrate with transparent ITO electrode, a shallow NV-diamond layer (100 nm depth), a 40 nm Au film serving as both optical filter and thermal strap, a 100 nm $SiO_2$ thermal isolation layer, and lithographically defined superconducting pixels with optional absorber material.

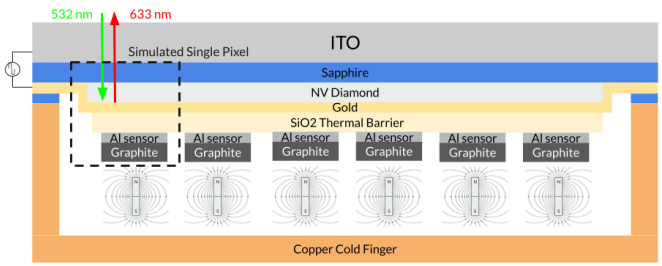


Fig. 1. Conceptual device stack and optical geometry showing an array of superconducting Al/graphite pixels with individual bias magnets on the copper cold finger. The dashed box indicates a single simulated pixel. Microwave drive for ODMR is capacitively coupled between the ITO and Au electrodes.

The NV ensemble is excited at 532 nm, and fluorescence near 633 nm is collected in an epifluorescence geometry through the same objective. A microwave field drives spin transitions for ODMR. The Au layer blocks residual pump light from reaching the superconductor while providing lateral thermal contact to the cold finger.

### B. Demonstrator Configuration

For proof-of-principle validation, we target a demonstrator pixel using 200 nm NbN ($T_c \simeq 15.7\,\mathrm{K}$) with a $1\,\mu$m Ge absorber on a $1.5 \times 5\,\mu\mathrm{m}^2$ footprint. The narrow width mitigates vortex nucleation under the $\sim$1 mT bias field required for ODMR operation [4]. Operating near $T_{\mathrm{op}} \approx 12\,\mathrm{K}$ provides thermal headroom while maintaining strong Meissner screening.

This configuration targets validation with 5.5 MeV alpha particles from $^{241}$Am, producing plateau durations $\tau \sim \mathcal{O}(10^2)\,\mu$s resolvable with standard photodetection electronics. Cross-talk between adjacent pixels is mitigated by the $1/r^3$ fall-off of the magnetic dipole field from each pixel and can be further suppressed through spatial deconvolution of the wide-field fluorescence image. The pixel pitch is chosen so that the field from a neighboring pixel at the NV layer is below the ODMR linewidth.

## III. Readout Principle and Simulation

### A. ODMR-Based Transduction

The NV ground-state spin Hamiltonian in an external magnetic field is

$$\mathcal{H} = D\,S_z^2 + E(S_x^2 - S_y^2) + \gamma_e \vec{B} \cdot \vec{S} \tag{1}$$

where $D \simeq 2877\,\mathrm{MHz}$ is the zero-field splitting at cryogenic temperatures, $E$ is the transverse strain parameter, and $\gamma_e \simeq 2.8\,\mathrm{MHz/G}$ is the electron gyromagnetic ratio [3], [5]. Under continuous-wave ODMR with a fixed microwave drive frequency $\nu_{\mathrm{MW}}$, the fluorescence exhibits a Lorentzian dip when $\nu_{\mathrm{MW}}$ matches a spin resonance.

We operate at $T_{\mathrm{op}} < T_c$ with $\nu_{\mathrm{MW}}$ tuned to the ODMR dip minimum. When energy deposition raises $T > T_c$, Meissner screening collapses, the local field at the NV layer increases, and the resonance detunes from $\nu_{\mathrm{MW}}$. The fluorescence rises from the ODMR dip to baseline, forming a plateau whose duration $\tau$ encodes the deposited energy.

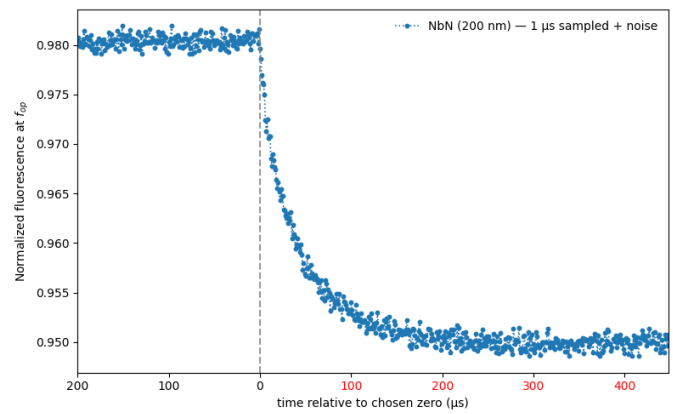


Fig. 2. Simulated NbN/Ge readout trace at $1\,\mu$s sampling with Gaussian fluorescence noise and $K_\uparrow = 3/K_\downarrow = 2$ trigger. The plateau remains clearly resolvable.

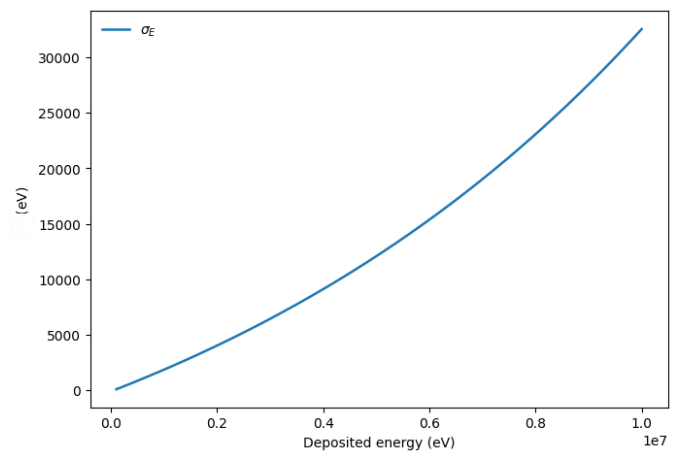


Fig. 3. Energy resolution for the NbN/Ge demonstrator at $T_{\mathrm{op}} \approx 12\,\mathrm{K}$. Sub-percent fractional resolution is predicted across the MeV region of interest.

### B. Energy Reconstruction via Timing

The transduction chain connects thermal response, London penetration depth, magnetic field at the NV center, and fluorescence intensity:

$$E_{\mathrm{dep}} \to T(t) \to \lambda(T) \to B_{\mathrm{NV}}(t) \to F(t) \to \tau(E_{\mathrm{dep}}) \tag{2}$$

where the London depth follows the Gorter–Casimir form $\lambda(T) = \lambda_0/\sqrt{1-(T/T_c)^4}$ [6].

Monte Carlo simulations including fluorescence intensity noise ($\sigma_F = 7.06 \times 10^{-4}$) [4], $1\,\mu$s sampling quantization, and random phase offsets demonstrate that plateau edges are resolvable with a trigger requiring $K = 3$ consecutive threshold crossings (Fig. 2). The energy resolution follows $\sigma_E \simeq |dE/d\tau|\,\sigma_\tau$, where $\sigma_\tau$ is the timing jitter (Fig. 3).

For the NbN/Ge demonstrator, simulations predict sub-percent fractional energy resolution ($\sigma_E/E < 1\%$) across the

MeV region of interest, with plateau durations of hundreds of microseconds.

## IV. Cryogenic NV-Diamond Characterization

To establish the operational envelope for our NV-diamond sensor, we characterize its cryogenic performance from 11 K to 200 K—demonstrating techniques and stability that directly translate to the final detector design. Measurements are performed on a commercial HPHT diamond (Adamas Nano, [NV] ∼2 ppm) with an ensemble NV layer, cooled in a closed-cycle cryostat (Lakeshore 335 temperature controller). ODMR spectra are acquired using a 532 nm diode laser, ADF4350 microwave synthesizer, and Thorlabs PM101 power meter.

### A. Zero-Field Splitting $D(T)$ and Strain $E(T)$

At zero applied magnetic field, the NV ground-state Hamiltonian (1) yields two ODMR transitions at frequencies $f_\pm = D \pm |E|$, where $|E| = \sqrt{E_x^2 + E_y^2}$ is the magnitude of the transverse strain [3]. We performed a temperature sweep from 11 K to 200 K with the permanent magnet retracted, recording ODMR spectra at each temperature stop.

Fig. 4(a) shows the two zero-field ODMR dip frequencies and their midpoint $D(T) = (f_+ + f_-)/2$. The zero-field splitting is approximately constant at $D \approx 2877.5$ MHz below 50 K, then decreases monotonically to $D \approx 2875.3$ MHz at 200 K. This behavior is well described by the two-phonon Bose–Einstein model of Cambria *et al.* [9]:

$$D(T) = D_0 + \sum_{i=1}^{2} \frac{c_i}{e^{\Delta_i/k_B T} - 1} \tag{3}$$

where $D_0 \approx 2877.7$ MHz is the zero-temperature value and the two phonon modes ($\Delta_1 \approx 59$ meV, $\Delta_2 \approx 146$ meV) account for the dominant electron-phonon (∼85%) and thermal expansion (∼15%) contributions [8]. At our operating temperature of 12 K, both phonon modes are exponentially frozen ($n(\Delta_1/k_B \cdot 12\,\mathrm{K}) \sim e^{-57} \approx 0$), making $D$ thermally stable to better than 10 kHz per 0.1 K—far below our 0.1 MHz bin resolution.

Fig. 4(b) shows the strain splitting $2E = f_+ - f_-$. The strain parameter increases from $E \approx 1.17$ MHz at 11 K to $E \approx 2.09$ MHz at 200 K. Unlike the axial parameter $D$, no standard analytic model for $E(T)$ exists in the literature; the only published quantitative result is a relative linear coefficient $dE/(E\,dT) = -1.4(3) \times 10^{-4}\,\mathrm{K}^{-1}$ near room temperature [**?**]. Our cryogenic data span a much wider range and show a monotonic increase, suggesting that differential thermal contraction between the diamond lattice and local defect structures progressively enhances the transverse crystal field at low temperatures. A phenomenological fit to these data will be reported in the final version.

### B. ODMR Contrast versus Temperature

The ODMR contrast—defined as the relative fluorescence dip depth—is a critical parameter for detector sensitivity, as it determines the signal amplitude available for plateau detection. We measured contrast as a function of temperature under two conditions: with a ∼43 G applied magnetic field (tracking individual Zeeman-split dips from each NV crystallographic axis), and at zero field.

**Fig. 4 — PLACEHOLDER**

*Top:* $D(T)$ with two-phonon Bose–Einstein fit (Eq. 3).
*Bottom:* $E(T)$ with phenomenological fit.
Both 11 K–200 K, zero-field data.

Fig. 4. (a) Zero-field ODMR dip frequencies $f_\pm = D \pm E$ and extracted $D(T)$ with the two-phonon Bose–Einstein fit of Eq. (3). (b) Strain splitting $2E$ and transverse strain parameter $E(T)$ versus temperature.

**Fig. 5 — PLACEHOLDER**

Contrast vs. temperature overlay:
$B = 0$ G (total contrast) and $B \approx 43$ G (per-axis).
Seven-level rate-model fit overlaid on both datasets.

Fig. 5. ODMR contrast versus temperature. (a) Individual dip contrast for all four NV crystallographic axes at $B \approx 43$ G. (b) Total zero-field ODMR contrast. Solid curves: seven-level rate-equation model [**?**] incorporating temperature-dependent ISC rates through the $^3E$ orbital branches and metastable singlet states.

Fig. 5(a) shows the contrast of all four NV-axis dips at $B \approx 43$ G from 11 K to 126 K. A striking non-monotonic behavior is observed: contrast decreases from ∼3% at 11 K to a minimum of ∼1.6% near 33 K, then recovers to a maximum of ∼3.8% near 72 K before declining again at higher temperatures. This behavior is reproduced across all four crystallographic orientations, confirming that it is an intrinsic property of the NV center rather than an experimental artifact. Fig. 5(b) shows the same non-monotonic trend at zero field over a broader temperature range (11–200 K), with contrast peaking near 75 K at ∼6.7%.

To quantitatively model this behavior, we employ a seven-level rate-equation model based on the framework of Rob-

**Fig. 6 — PLACEHOLDER**

Three-panel ODMR spectra at 12 K:
(a) $B = 0\,\mathrm{G}$ (b) $B \approx 46\,\mathrm{G}$ (c) $B \approx 210\,\mathrm{G}$
Data (points) with Hamiltonian+Lorentzian+seven-level fit (curves).

Fig. 6. Representative ODMR spectra at 12 K at three applied fields, with Hamiltonian-driven Lorentzian fits (Eq. **??**). (a) Zero field: two dips at $D \pm E$. (b) $B \approx 46\,\mathrm{G}$: eight resolved dips from four NV orientations. (c) $B \approx 210\,\mathrm{G}$: widely separated dips with per-axis contrast modulated by the seven-level model.

ledo *et al.* [**?**], which tracks the populations of the ground-state ($m_s = 0$ and $m_s = \pm 1$), excited-state ($^3E$ orbital branches $E_x$, $E_y$ for each spin projection), and the two singlet levels ($^1A_1$ and $^1E$). The temperature dependence enters through (i) phonon-mediated hopping between the $E_x$ and $E_y$ branches of $^3E$, which modifies the spin-selective intersystem crossing (ISC) rates [3], and (ii) the temperature-dependent singlet lifetime $\tau_{\mathrm{singlet}}(T) = [\tau_0^{-1} + A\, e^{-\Delta_s/k_B T}]^{-1}$ with $\Delta_s \approx 16.6\,\mathrm{meV}$ [**?**]. The non-monotonic contrast arises from the competition between these orbital-mixing and thermal-depolarization effects. The contrast minimum near 33 K represents a sensitivity dead zone that should be avoided in the choice of operating temperature. Our demonstrator operating point ($T_{\mathrm{op}} \approx 12\,\mathrm{K}$) lies in a region of adequate contrast ($\sim$3%), though the global maximum near 72 K may motivate future designs using superconductors with higher $T_c$ values.

### C. Peak Fitting and Multi-Axis Zeeman Spectroscopy

We fit the full ODMR spectrum at each field and temperature using a combined model: the Hamiltonian-derived peak positions from Eq. (4) determine the Lorentzian center frequencies, while the seven-level rate-equation model described in Sec. IV-B supplies the predicted contrast for each dip as a function of temperature. The complete lineshape model is

$$P(f) = P_0 \left[ 1 - \sum_{i=1}^{N_{\mathrm{axes}}} \sum_{s=\pm} C_{s,i}(T)\, \mathcal{L}(f;\, f_{s,i},\, \Gamma_i) \right] \tag{4}$$

where $\mathcal{L}$ is a Lorentzian with FWHM $\Gamma_i$, $f_{s,i}$ are the transition frequencies from Eq. (4), and $C_{s,i}(T)$ is the per-dip contrast from the rate model. At zero field ($N_{\mathrm{axes}} = 1$, two transitions), the model reduces to the "W-shape" doublet. At high field ($N_{\mathrm{axes}} = 4$, eight transitions), the fit extracts the full field vector.

The diamond lattice supports four equivalent NV orientations along $\langle 111 \rangle$ crystallographic directions. An external magnetic field $\vec{B}$ projects differently onto each axis $\hat{n}_i$, producing up to eight distinct ODMR dips. In the perturbative regime ($\gamma_e |\vec{B}| \ll D$), the transition frequencies for axis $i$ are [5]:

$$f_{\pm,i} = D + \frac{\gamma_e^2 B_{\perp,i}^2}{2D} \pm \sqrt{(\gamma_e B_{\parallel,i})^2 + E^2} \tag{5}$$

where $B_{\parallel,i} = \vec{B} \cdot \hat{n}_i$ and $B_{\perp,i} = \sqrt{|\vec{B}|^2 - B_{\parallel,i}^2}$. We performed calibrated field sweeps from 10 G to 36 G at 11 K. All eight dips become individually resolved above $\sim$30 G (inter-dip gap $>$ 10 MHz FWHM linewidth). A global fit of (4) to all four dip pairs at high field yields the full field vector: $|\vec{B}| = 48.4\,\mathrm{G}$ with polar angles $\theta_A = 47.8^\circ$ and $\theta_B = 64.4^\circ$ relative to the two inequivalent axis groups, and a common strain parameter $E = -6.1\,\mathrm{MHz}$. From the four measured splittings, the Cartesian field components can be reconstructed via the inverse projection $B_k = (\sqrt{3}/4) \sum_i s_i (\hat{n}_i)_k \cdot B_{\parallel,i}$, where $s_i = \pm 1$ is resolved by the known magnet geometry. This intrinsic vector magnetometry capability provides an absolute field calibration at each pixel without external reference.

### D. Spin-Lattice Relaxation Time $T_1$

Spin-lattice relaxation ($T_1$) measurements were performed at $B \approx 46\,\mathrm{G}$ using a three-phase pulsed protocol: baseline measurement (MW off), spin driving (MW on, 500 ms), and recovery monitoring (MW off, 1.2 s). A gold-standard 2-parameter exponential fit with the baseline pinned to the measured equilibrium value extracts $T_1$ from the recovery curve at each temperature (Fig. 6).

Below $\sim$75 K, $T_{1,\mathrm{off}} \approx 250\,\mathrm{ms}$ remains approximately constant, indicating a cross-relaxation-limited regime dominated by dipolar coupling to substitutional nitrogen (P1 centers) and NV–NV interactions, both of which are temperature-independent [10]. Above 75 K, $T_1$ decreases to $\sim$195 ms at 120 K, consistent with the onset of the Orbach two-phonon relaxation process (virtual excitation through $^3E$) with an activation energy $\Delta \approx 42\,\mathrm{meV}$. A control experiment with the microwave drive disabled confirmed that the recovery signal is genuine spin physics (70:1 signal-to-drift ratio). These long $T_1$ times ($\gg$ the $\sim$100 $\mu$s plateau durations expected from the demonstrator) confirm that spin relaxation does not limit the readout bandwidth.

## V. Multi-Modal Sensing Capability

The cryogenic optical platform developed for NV-diamond characterization has proven versatile beyond magnetometry. During a routine cooldown in which the vacuum chamber was briefly exposed to atmospheric pressure at 11.4 K ($\sim 10^{-8}$ Torr base), a thick layer of amorphous solid water (ASW) condensed on the cold stage along with trapped atmospheric gases ($N_2$, $O_2$, Ar). The 532 nm excitation laser simultaneously probes both NV fluorescence and Raman-active vibrational modes. The O–H stretch Raman band ($\sim$3200 $\mathrm{cm}^{-1}$ shift, emitted at $\sim$641 nm) falls within the passband of the collection dichroic filter and was monitored continuously during warmup

**Fig. 7 — PLACEHOLDER**

$T_1$ vs. temperature at $B \approx 46\,\mathrm{G}$.
Orbach fit: $1/T_1 = 1/T_{1,0} + A\,e^{-\Delta/k_B T}$.
Ensemble-averaged recovery curves at selected $T$.

Fig. 7. Spin-lattice relaxation time $T_{1,\mathrm{off}}$ versus temperature at $B \approx 46\,\mathrm{G}$. Below $\sim$75 K, $T_1$ plateaus at $\sim$250 ms (cross-relaxation limited by dipolar coupling to P1 centers [10]). Solid curve: Orbach fit with activation energy $\Delta \approx 42\,\mathrm{meV}$.

**Fig. 8 — PLACEHOLDER**

Integrated O–H Raman intensity vs. temperature (11–220 K).
Color-coded regions I–V: HDA $\rightarrow$ LDA $\rightarrow$ collapse $\rightarrow$ stable $\rightarrow$ crystallization $\rightarrow$ sublimation.

Fig. 8. Integrated O–H stretch Raman intensity versus temperature during warmup, with five structural regimes color-coded. The $\sim$4 Hz sampling rate resolves all transitions in real time.

at $\sim$4 Hz temporal resolution using the broadband power meter [11].

Five distinct structural regimes were resolved in the integrated Raman intensity as a function of temperature:

**(I) Stable high-density amorphous (HDA) ice** below 23 K: constant baseline intensity from a microporous, highly disordered ice film.

**(II) HDA$\rightarrow$LDA transition** (23–54 K): the Raman intensity increased by a factor of $2.08\times$, concurrent with explosive trapped-gas desorption ("volcano desorption") that drove the chamber pressure from $\sim 10^{-8}$ Torr to $>$50 mTorr, triggering ion gauge overpressure protection. The intensity rise reflects pore network opening and hydrogen bond reorganization during the amorphous-to-amorphous restructuring.

**(III) Pore collapse and densification** (54–60 K): a rapid 60% intensity crash as the remaining microporous structure collapses into compact LDA ice, reducing the scattering volume and optical path length. The signal stabilized near 110 K at $0.71\times$ baseline.

**(IV) Crystallization** (155–190 K): a broad 14% intensity recovery as LDA ice crystallizes to stacking-disordered ice ($\mathrm{I_{sd}}$) and subsequently converts to hexagonal ice ($\mathrm{I_h}$). The spectral narrowing of the O–H band in the crystalline phase concentrates Raman intensity into the detection window.

**(V) Sublimation** ($>$200 K): progressive intensity decline as the ice film thins.

All observed transition temperatures agree with published values from infrared spectroscopy and electron diffraction. These observations demonstrate that the NV-diamond cryogenic platform enables real-time, *in situ* broadband Raman spectroscopy of surface adsorbates at temporal resolution three orders of magnitude higher than conventional dispersive Raman—a capability that complements its primary magnetometry function and may find application in cryogenic surface science, thin-film characterization, and astro-ice analog studies.

## VI. Roadmap and Outlook

We have defined a staged validation pathway:

**D0 (Room-temperature benchtop):** Demonstrate calibrated $\Delta B$ detection using wide-field NV magnetometry with lock-in ODMR readout. Gate criterion: SNR $\geq 5$ at design dwell time.

**D1 (Coupling NbN thin films with NV centers):** Integrate NbN thin films with NV diamond at $T \approx 12\,\mathrm{K}$. Validate heater-step calibration and extract $\partial B/\partial E$. Gate: reconstruct injected energy within 20%.

**D2 (Mini-array):** Fabricate small NbN/Ge pixels on preformed NV-center heterostructures and quantify crosstalk via cross-correlation. Gate: nearest-neighbor crosstalk $<$5%.

**D3 (Alpha validation):** Expose demonstrator to $^{241}$Am alphas and verify energy reconstruction against the known 5.5 MeV line.

Beyond the MeV demonstrator, the architecture scales to sub-eV detection by reducing pixel heat capacity. Simulations of Al(20 nm)/graphite(200 nm) pixels at $T_c \simeq 1.1\,\mathrm{K}$ predict thresholds near 0.75 eV with $\sigma_E < 0.4\,\mathrm{eV}$—relevant for searches for light dark matter candidates including dark photons and axion-like particles [7]. The optical readout enables scaling to $\sim 10^9$ pixels via wide-field imaging of NV fluorescence, a density impractical with electrical readout. A $10\,\mathrm{cm}^2$ detector with $1\,\mu\mathrm{m}^2$ pixels at $2\,\mu$m pitch would contain $2.5\times 10^9$ pixels—each independently read out through a single imaging objective with no electrical connections to the cold stage.

Our CW-ODMR characterization establishes a magnetic field sensitivity of $\eta \approx 14\,\mu\mathrm{T}/\sqrt{\mathrm{Hz}}$ at the drift floor, achieved with 100-pair averaging at 20 ms dwell per frequency bin. This sensitivity, combined with the measured ODMR linewidth ($\sim$10 MHz FWHM) and contrast ($>$2% at the optimal operating temperature), provides the quantitative inputs for modeling

the minimum detectable field change $\delta B$ during a Meissner screening event.

The cryogenic characterization presented here identifies $T_{\mathrm{op}} \approx 12\,\mathrm{K}$ as a suitable operating point, offering adequate ODMR contrast ($\sim$3%), long spin relaxation ($T_1 > 200\,\mathrm{ms}$), and well-resolved Zeeman structure. The measured $D(T)$ and $E(T)$ data provide the quantitative inputs needed for full Hamiltonian modeling of the NV response under varying thermal and magnetic conditions during detector operation.

## VII. Conclusion

We have presented a bolometric architecture combining superconducting Meissner screening with NV-center optical magnetometry, supported by comprehensive cryogenic characterization of the diamond sensing element. Zero-field measurements from 11 K to 200 K establish $D(T)$ and $E(T)$, revealing a temperature-dependent strain parameter that grows from 1.2 MHz to 2.1 MHz. ODMR contrast exhibits a non-monotonic temperature dependence with a minimum near 33 K and maximum near 72 K, observed consistently across all four NV crystallographic orientations. Spin relaxation times exceed 200 ms below 80 K, confirming that $T_1$ does not limit readout bandwidth.

These measurements, combined with multi-axis Zeeman spectroscopy and the serendipitous demonstration of real-time Raman monitoring of amorphous ice phase transitions, validate the cryogenic NV-diamond platform as a versatile sensing tool. The NbN/Ge demonstrator design, with simulated sub-percent energy resolution at MeV scales, provides a concrete pathway toward sub-eV threshold particle detection for next-generation quantum sensors.

## Acknowledgments

This work was supported by the University of Texas at Arlington Department of Physics.

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